Econ 2 - Lecture 8 - 2/4/26 (Weekly Coffee Hours ↳ 2/5 @ 1PM)

Midterm Avg = 21.63/30 (Spring 25 Avg. 21.75/30)

Note: 30% of grade = quizzes, discussion, participation

$2^{nd}$ Half of Quarter is more complex/graphical ⇒ Macro Equilibrium

No lecture quiz this week (week 5)

Today: Transition to Theoretical Model

Goal: Explain why the level of GDP (Y) is at a particular amount

Why is Y = 31T (Nominal), 24T (Real)

Bigger Goal: full employment level of $Y = \bar{Y}$ (Y-bar)
↳ 0% cyclical unemployment

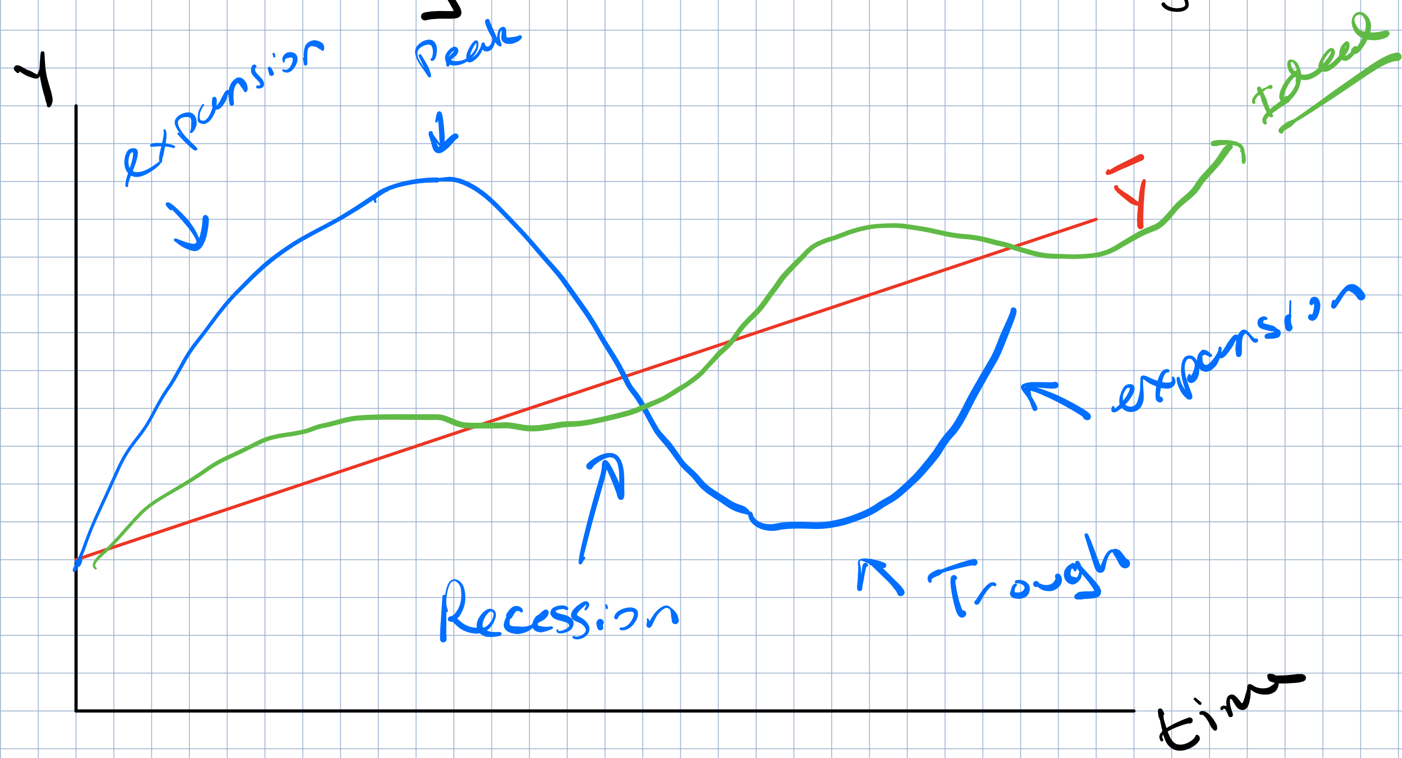
1800s: Uncomplicated growth
Classical Economists: "Laissez-faire" = "Let it be"

Great Depression: ~15 years to fully recover

$\underbrace{\text{John Maynard Keynes}}_{\text{Policy Intervention} \Rightarrow \text{Priority}}$ : "In long-run, we're all dead."

Model → Macroeconomy, Keynesian Approach

# Macroeconomy follows the business cycle




Dating Recessions: National Bureau of Economic Research (NBER)

Recession: Multiple quarters of decreased $\underbrace{\text{economic activity}}_{\text{Production, jobs}}$

Depression: 3+ year recession → +10% reduction in GDP

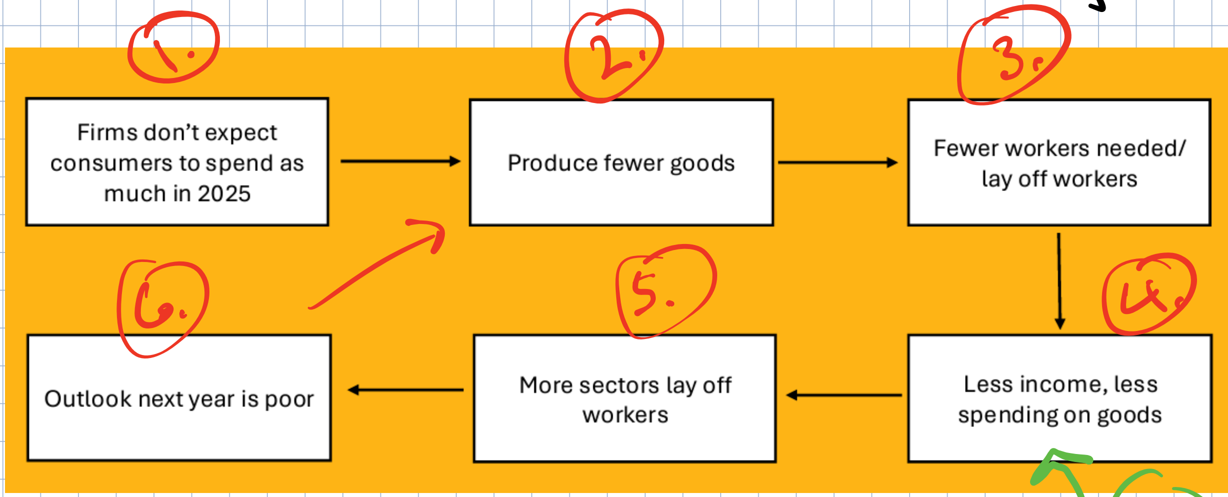
UE in 1930s ≈ 19 - 25%

Great Recession: Recession that lasted 18 months

→ Avg. Recession ≈ 9 months

→ Less than 10% drop in GDP

# How do we model the entire economy?




C

Households are concerned about the future (A) → Saving a lot (B)

## Assumptions being made?

More spending ⇒ More income

⇑ ⇗

More production

If Spending = $\phi$, Production = $\phi$ ⇒ Income = $\phi$

## Guiding Equilibrium Principle

$\underbrace{Y = \text{Output} = \text{Production} = \text{Income}}_{\text{GDP}} = \text{Spending}$

$$\underbrace{\text{Equilibrium}}_{Y = AE} \text{ is at } Y = \underbrace{\text{Aggregate Expenditures (AE)}}_{\text{Spending}}$$

Goal: find the level of $Y$ where $Y = AE$

Bigger Goal: find our full-employment level of $Y \Rightarrow \bar{Y} = AE$

Understand what goes into spending?

Define Aggregate Expenditures (AE)

Biggest spending group = Households = Consumption (C)

↳ Account for ~70% of total spending (AE)

What determines household spending?

Biggest determinant of C = Income (Y) in short-run

feels circular:

→ Find where $Y = AE$

→ 70% of AE is from C

→ Biggest driver of C is Y

1.) Income = Y $\Rightarrow$ $\underbrace{\text{Disposable Income}}$ (DI)

DI = Income (Y) - Taxes (T) = Take-Home Income

If DI ↑ → Y ↑ → C ↑
→ T ↓ → C ↑

More DI
$\Rightarrow$ More C

If DI ↓ → Y ↓ → C ↓
→ T ↑ → C ↓

2.) Wealth = Value of what we own - Value of what we owe
↓
Equity

Assets
- Home
- Stocks
- Bonds
- Crypto?
- Cash
- Gold

Liabilities
- Mortgage
- Student Loans
- Car Loan
- Credit Card

As wealth ↑ $\Rightarrow$ C ↑

As wealth ↓ $\Rightarrow$ C ↓

3.) Interest Rates (r)

- → Cost of borrowing, reward for saving
- → Related to prices (week 9)

As $r \uparrow \Rightarrow C \downarrow$

As $r \downarrow \Rightarrow C \uparrow$

4.) Expectations: What does next year look like?

5.) Preferences: Kids, family size, Pandemic Lockdown